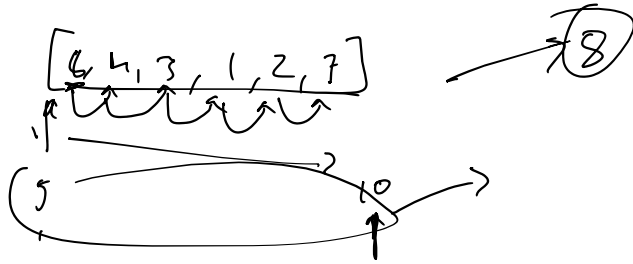
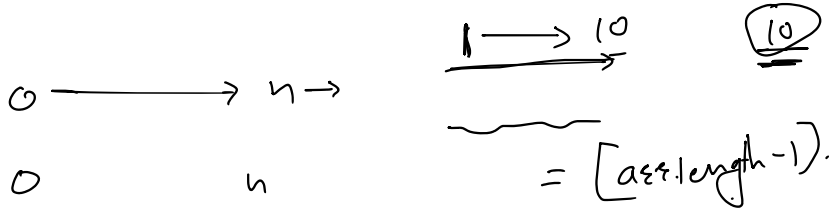


Time complexity & space complexity



Asymptotic notation & Experimental way

① worst case

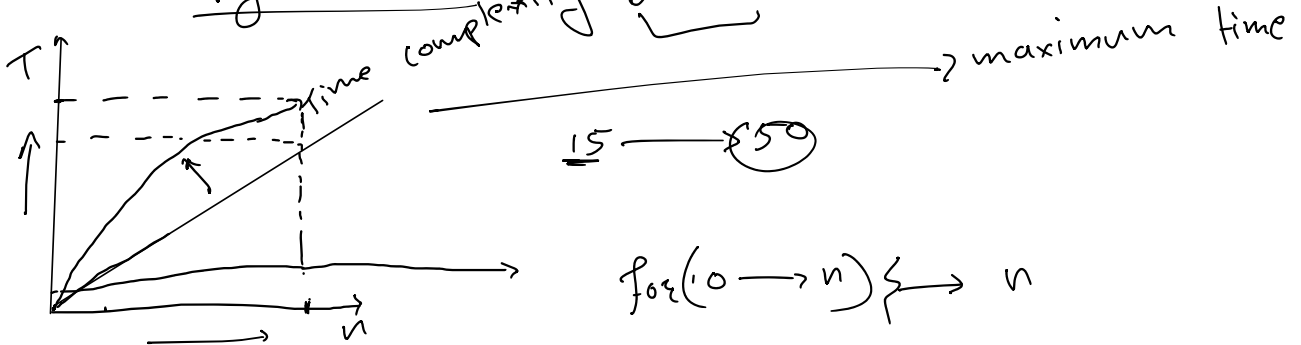
① Best case

③ Average

Big O notation $\longrightarrow$ upper bound

0 $\longrightarrow$ n $\longrightarrow$ 50

15 $\longrightarrow$ 50



$O(1) \longrightarrow$ constant

$O(n^2)$

for (0 $\longrightarrow$ n) $\{ \longrightarrow$ n

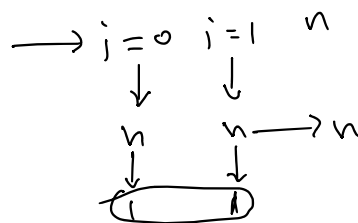
for (0 $\longrightarrow$ n) $\{$
 $\quad$ sort()
 $\quad \}$

$\}$

$$\frac{n \times n}{n^2}$$

$$f(n) = \cancel{an^2}b + n + c$$

$$\frac{\cancel{n^2} + n}{n^2}$$

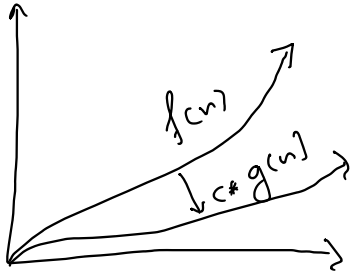


$$\frac{(n \times n)}{n^2}$$



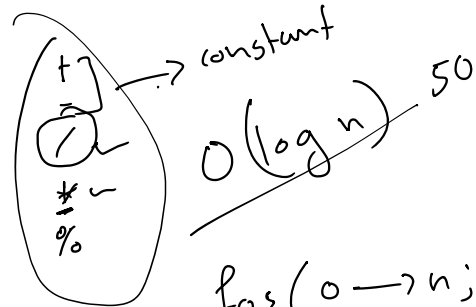
Omega $\rightarrow \Omega(1), \Omega(n), \Omega(n^2)$

$\rightarrow$ lower bound

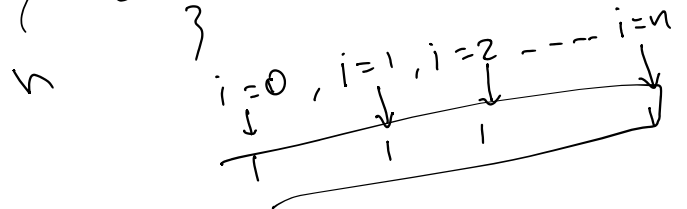


$\rightarrow$ minimum time

- $O(1)$ ✓
- $O(\log n)$
- $O(n \log n)$
- $O(n)$
- $O(n^2)$
- $O(n^3)$



for (0 $\rightarrow$ n; i++)



for (i=50; i>0; i/=2)

$$\begin{aligned}
 i &= 50 \\
 25 &= 50/2 \\
 12 &= 25/2 \\
 6 &= 12/2 \\
 3 &= 6/2 \\
 1 &= 3/2 \\
 0 &= 1/2
 \end{aligned}$$

1000000

`for(0 → n; i++) {`

`for(0 → n; i *= 2) {`
`}`

`}`

$(n \log n)$

$n \times \log n$
 $O(n \log n)$

```
for (i = 1; i <= n; i++) {
    for (int j = 1; j <= i * i; j++) {
        for (k = 1; k <= n / 2; k++) {
            // O(n^4)
            System.out.println("hey");
        }
    }
}
```

$n \times n^2 \times n$
 $\underline{\hspace{1cm}}$
 (n^4)

```
for (i = n / 2; i <= n; i++) {
    for (int j = 1; j <= n / 2; j++) {
        for (k = 1; k <= n; k = k * 2) {
            System.out.println("hey");
        }
    }
}
```

$n \times n \times \log n$

$n^2 \log n$

$n = 20$

```
for (i = 1; i * i <= n; i++) {
    System.out.println("hey");
}
```

$i = 1 \quad i^2 = 1$

```
for (i = 1; i * i <= n; i++) {
    System.out.println("hey");
}
```

$n = 20$

$i = 1, i = 2$
↓ ↓
1 1

$i = 3, i = 4$
↓ ↓
1 1

$i = 5$

$\left\{ \begin{array}{l} 4 < 20 \\ 9 < 20 \\ 16 < 20 \\ 25 < 20 \end{array} \right.$

space complexity → $\left\{ \begin{array}{l} \text{heap} \rightarrow \text{object/array} \\ \text{stack} \rightarrow \text{function} \end{array} \right.$

$\left\{ \begin{array}{l} a = 10 \\ b = 20 \end{array} \right.$

$\leftarrow \begin{array}{l} \text{temp} = a \\ a = b \\ b = \text{temp} \end{array} \rightarrow O(1)$
↑

$\text{arr} = \text{new int}[n]$ → $O(n)$
(5)